Yash Golani

SOFTWARE SYSTEMS · APPLIED AI · ROBOTICS

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Software engineer building distributed video-processing systems, computer-vision pipelines, and full-stack applications. Experience connecting inference services, data pipelines, APIs, and investigation interfaces, with a focus on recoverable workflows, reliable processing, and maintainable service integration.

PROFESSIONAL EXPERIENCE

ESSI Integrated Technologies

New Delhi, India

Project Engineer, R&D Software Systems

Apr 2026–Present · Full-time

Systems Engineering Intern | Jun 2025–Mar 2026

- **Distributed processing:** Built Python/FastAPI orchestration for live and recorded video across inference servers, using Kafka and Redis for service coordination. Implemented task dispatch, retries, recovery, camera assignments, and shared GPU processing.
- **Computer vision runtime:** Developed multi-camera analytics with OpenCV, YOLO, InsightFace, and ONNX Runtime for vehicles, plates, faces, crowds, and traffic. Addressed worker isolation, stream reconnection, camera lifecycle management, and consistent event outputs.
- **Full-stack applications:** Built React/TypeScript interfaces and FastAPI services backed by PostgreSQL/pgvector for alerts, historical search, maps, and configuration. Connected sightings into investigation timelines, routes, associations, and evidence exports.
- **Model lifecycle & video retrieval:** Developed annotation, training, evaluation, model-versioning, and export workflows. Built video-understanding pipelines using VLM captions, SentenceTransformers embeddings, Qdrant similarity search, and retrieval-augmented generation.
- **Systems integration:** Contributed authenticated APIs, evidence playback, and alerts across VA, VMS, and ICCC applications. Developed radar tracking, camera calibration, and PTZ workflows linking sensor observations to operator review.

TECHNICAL SKILLS

Languages & interfaces: Python, TypeScript, JavaScript, C++, React, FastAPI, Flask, REST, WebSockets

Systems & data: PostgreSQL, pgvector, Kafka, Redis, Docker, Git, Qdrant, event-driven processing

AI & algorithms: PyTorch, OpenCV, YOLO, ONNX Runtime, TensorFlow, scikit-learn, VLMs, RAG, LoRA, LSTM, FFT/STFT, XGBoost, pose geometry, inverse kinematics

Robotics & embedded: ROS 2, RViz2, Gazebo, ESP32, Arduino, Embedded C/C++

SELECTED PROJECTS

Jarvis: Desktop Agent

GitHub ↗

Connected voice/text interaction, tool orchestration, browser automation, and permission-gated actions through a desktop interface and backend services.

React · Tauri · FastAPI · Python · Ollama · Playwright

Traffic VLM: Video Understanding & Retrieval

GitHub ↗

Combined Qwen2.5-VL/LLaVA captioning, YOLO-based plate analysis, sentence-embedding retrieval, and LoRA training workflows in a local Streamlit application.

PyTorch · Transformers / PEFT · SentenceTransformers · Ollama · Streamlit

TurtleBot 4: Perception

GitHub ↗

Integrated ROS 2 image processing with YOLOv8 detection, confidence/class filtering, and OpenCV overlays for robot perception.

Python · ROS 2 · YOLOv8 · OpenCV · PyTorch

EDUCATION

B.Tech, Robotics and Automation

2022–2026

Symbiosis Institute of Technology, Pune | CGPA: 7.51